



Problem Formulation

Classification and Regression

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Problem Formulation

Outline

Notation
Convention

Problem
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Outline

① Notation Convention

② Problem Formulation



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Notation Convention

Notation

Used Throughout This Course

Vectors & matrices

- **Vectors** (column): $\mathbf{x}$, $\mathbf{w}$, $\mathbf{b}$ — lowercase bold
- **Matrices/tensors**: $\mathbf{X}$, $\mathbf{W}$, $\mathbf{Z}$ — uppercase bold

Batch convention

Stack $\mathbf{x}_i^T$ as rows. Shapes: tabular $\mathbf{X} \in \mathbb{R}^{B \times D}$; image (B, C, H, W) ; sequence (B, T) or (B, T, D) .

Output & labels

Model output: $\hat{\mathbf{Y}}$ or $\hat{y}$; ground truth: $\mathbf{Y}$ or y ; logits: $\mathbf{Z}$; probabilities: $\hat{\mathbf{P}}$.

Parameters & sizes

θ (all params); $\mathbf{W}$, $\mathbf{b}$ (weights, bias). B batch size; D features; K classes; N total samples.

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What is a Machine Learning Problem?

General Setting

- Given a dataset of input–output pairs:

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$$

- Learn a function $f(\cdot)$ such that:

$$f(\mathbf{x}) \approx y$$

- The function f is parameterized and learned from data

Machine learning is about learning a mapping from data, not writing explicit rules.



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Supervised Learning

Supervised Learning

- Training data includes:
 - Input features $\mathbf{x}$
 - Ground-truth label or target y
- Goal: learn a function

$$f : \mathbf{x} \mapsto y$$

that generalizes to unseen data

Two Main Problem Types

- **Classification**
- **Regression**



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Classification

Classification Problem

- Output y belongs to a **finite set of classes**

$$y \in \{1, 2, \dots, K\}$$

- Model outputs class probabilities:

$$P(y = k \mid \mathbf{x})$$

Common Classification Types

- Binary classification
- Single-label multi-class classification
- Multi-label classification



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Regression

Regression Problem

- Output y is a **continuous value**

$$y \in \mathbb{R} \quad \text{or} \quad \mathbf{y} \in \mathbb{R}^K$$

- Model predicts a real-valued quantity:

$$\hat{y} = f(\mathbf{x})$$

Typical Examples

- Price prediction
- Temperature forecasting
- Signal reconstruction



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Notation and Batch Convention

Vector and Matrix Convention

- All vectors are **column vectors**
- A batch is formed by stacking $\mathbf{x}^T$ as rows

Notation

- Input batch:

$$\mathbf{X} \in \mathbb{R}^{B \times D}$$

(row n is $\mathbf{x}_n^T$)

- Output batch:

$$\mathbf{Y} \in \mathbb{R}^{B \times K}$$

- B : batch size, D : input dimension, K : output dimension (or classes)



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Learning Objective

General Learning Objective

- Learn parameters θ of a model f_θ
- Minimize a loss function:

$$\theta^* = \arg \min_{\theta} \mathcal{L}(f_\theta(\mathbf{X}), \mathbf{Y})$$

Different problems use different loss functions, but share the same learning principle.



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Data Distribution and Generalization

Underlying Data Distribution

- Data are assumed to be sampled from an unknown distribution:

$$(\mathbf{x}, y) \sim \mathcal{P}(\mathbf{x}, y)$$

- True goal: minimize **expected risk**

$$\mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{P}} [\ell(f_{\theta}(\mathbf{x}), y)]$$

What We Actually Minimize

- Using finite training set:

$$\frac{1}{N} \sum_{i=1}^N \ell(f_{\theta}(\mathbf{x}_i), y_i)$$

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Training, Validation, and Test Sets



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Dataset Splitting

- **Training set:** learn model parameters
- **Validation set:** tune hyperparameters, early stopping
- **Test set:** final unbiased evaluation

Important Rule

- Test data must NOT be used during training or tuning
- Avoid **data leakage**

Test set should be touched only once.

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Output Space of Learning Tasks

Regression

- Scalar output: $y \in \mathbb{R}$
- Vector output: $\mathbf{y} \in \mathbb{R}^K$

Classification

- Binary: $y \in \{0, 1\}$
- Multi-class (single-label): $y \in \{1, \dots, K\}$
- Multi-label: $\mathbf{y} \in \{0, 1\}^K$

The output space determines activation functions and loss functions.



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Parametric Models and Inductive Bias



Parametric Model

- Model is defined as:

$$f_{\theta}(\mathbf{x})$$

- θ : learnable parameters (weights $\mathbf{W}$, biases $\mathbf{b}$)

Inductive Bias

- Built-in assumptions about the solution
- Examples:
 - Linear models assume linear relationships
 - CNN assumes locality and translation invariance
 - Transformer assumes token interactions via attention

Learning is impossible without some form of inductive bias.

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Examples of Loss Functions

Regression

$$\ell(y, \hat{y}) = (y - \hat{y})^2 \quad (\text{Mean Squared Error})$$

Classification

- Binary Cross-Entropy
- Categorical Cross-Entropy

Loss function defines what it means to make a good prediction.



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